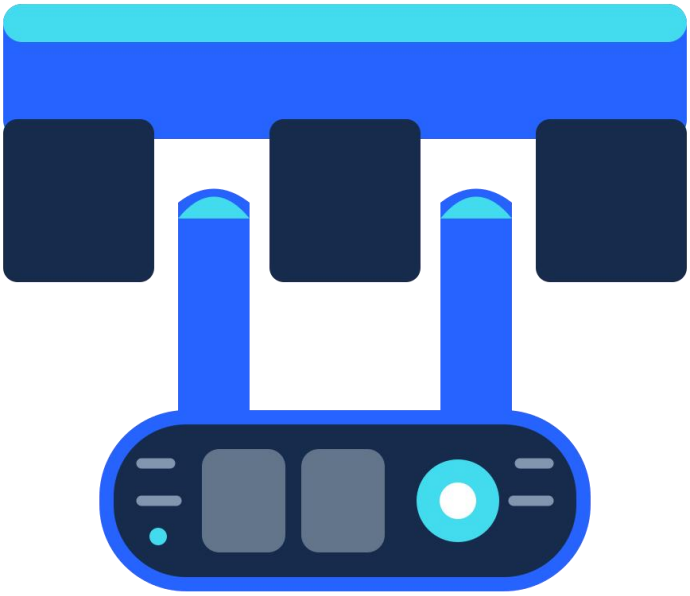
MRDVS



AW3 Smart Docking

Smart Docking User Manual

Standard international offering: pure 3D geometric positioning
2D semantic positioning is not included

Zhejiang MRDVS Technology Co., Ltd.

V0.2 · 2026-10-30

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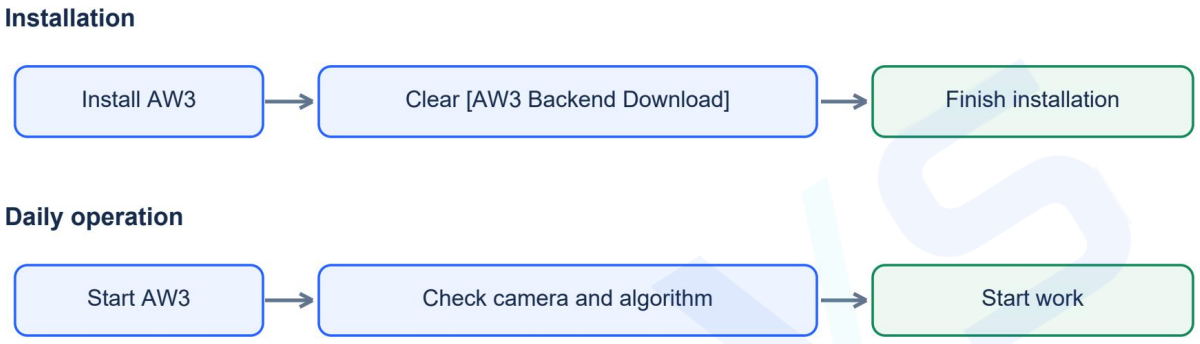
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1 Product Overview

Smart Docking runs on the AW3 algorithm platform and supports 2D semantic and pure 3D positioning. It outputs the pallet fork-entry reference position, orientation and positioning status for autonomous forklift docking.

1.1 Operating Prerequisites

1. Install AW3. No AW3 backend is required; clear the [AW3 Backend Download] checkbox.
2. Connect the camera and select the [Pallet Docking] task in Camera Groups.
3. Check that images and point clouds update continuously and the algorithm runs normally.



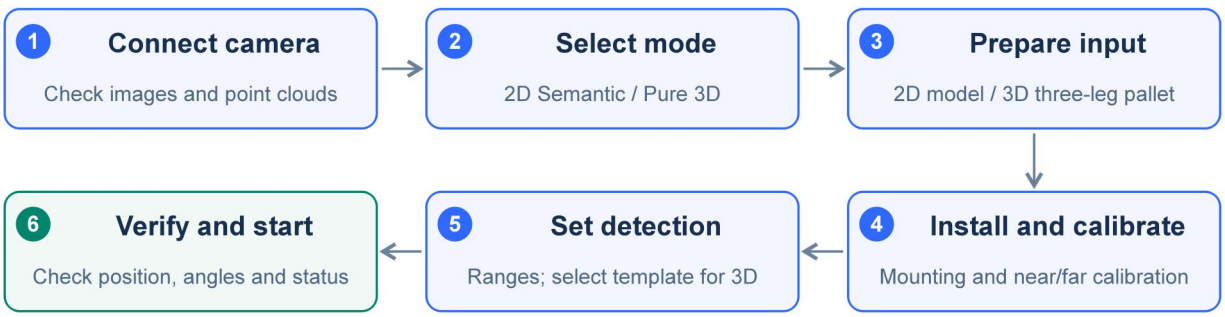
AW3 installation and daily startup: select and run Smart Docking in the platform

Streaming errors If AW3 streaming fails after configuring a camera in LxCameraViewer, open [Device Connection / Base Camera] > [Central Service] > [Restart current center]. No restart is needed if streaming is normal.

For instructions on all AW3 platform functions and firmware upgrades, refer to the AW3 User Manual.

1.2 First-Use Workflow

Before deployment: collect site-pallet RGB images for 2D model training and validation. 3D requires no training-data preparation.



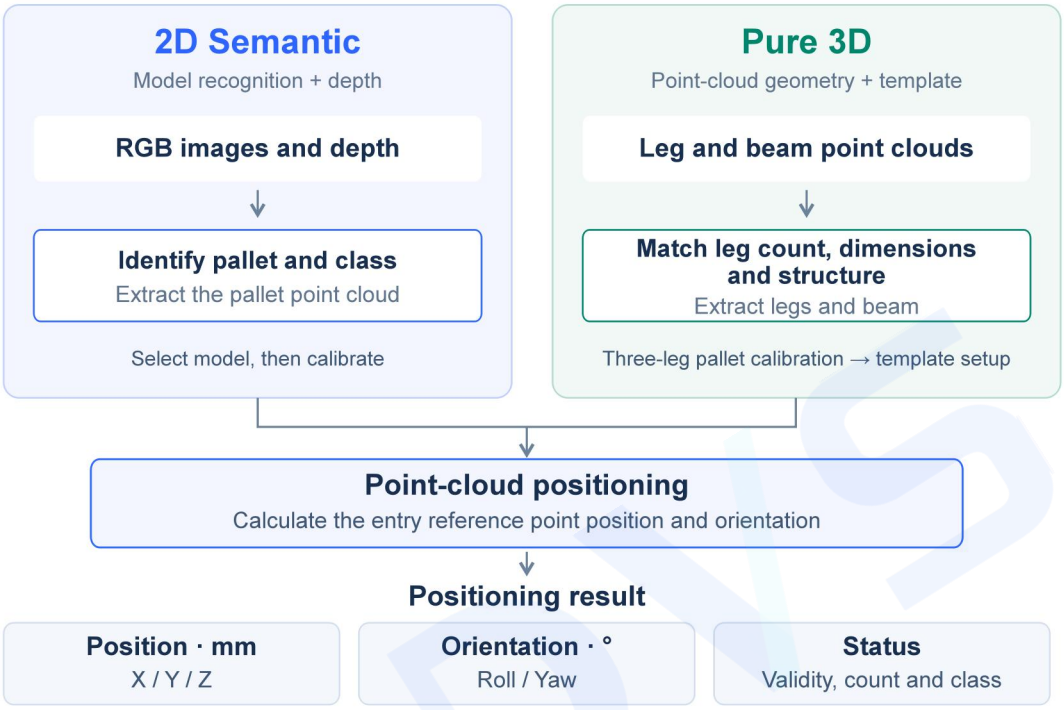
Before starting: secure the camera; keep the entry face and floor visible; confirm pallet dimensions and camera working distance suit the task.

Check the conditions in Section 2.2 before use. For operating steps, see Section 3.2 for 2D or Section 3.3 for 3D.

2 Before You Start

2.1 Choose a Positioning Mode

The standard international offering uses pure 3D geometric positioning. It does not include 2D semantic positioning.



2D semantic positioning and pure 3D positioning

The AW3 JSON positioning result provides five valid axes: X, Y, Z, Roll and Yaw. Position is in mm and angles are in °; Pitch is not a valid output. Load status applies only to pure 3D mode with load detection enabled. Validate results before vehicle use.

Function	2D Semantic Positioning	Pure 3D Positioning
2-, 3- and 4-leg pallets	Supported with a suitable model	Supported with a geometry template
With / without lower beam	Classification supported	Not classified
Load detection	Not supported	Supported when enabled

International projects: configure pure 3D in Section 3.3, then verify results in Section 3.4. Use Section 3.2 only for a delivered 2D semantic solution.

Solution and Computing Options

The standard international offering uses pure 3D geometric positioning. It does not include 2D semantic positioning.

For both S11 2D and pure 3D, use the customer’ s existing AGV industrial PC or computing platform first. No dedicated board is needed if it meets the requirements of the selected mode.

S11 · 3D | Customer AGV compute (international recommendation)

Pure 3D geometry · Runs on the customer’s Linux industrial PC

Suitable PC: no extra board. Optional board after assessment and agreement, at extra cost.

M4 Mega | Onboard compute

Pure 3D geometry · Runs inside the camera

No external computing device required

S11 · 2D | Customer AGV compute (not standard internationally)

Customer platform must use RK3576 / RK3588; site RGB training data required

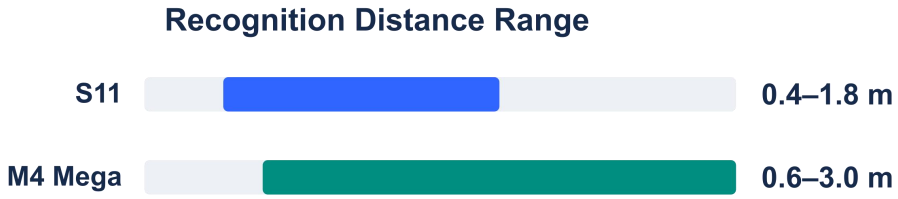
Insufficient compute: optional board after assessment and agreement, at extra cost.

S11 compute: 2D requires RK3576/RK3588; pure 3D recommends Intel Core i5-class or RK3568 and above. Linux and algorithm compatibility must be confirmed; lower 3D specifications require assessment. CPU/GPU usage affects processing time. For either mode, insufficient capacity may be addressed with an additional compute board, subject to assessment and agreement, at extra cost.

AW3 PC: Windows, for configuration, monitoring and troubleshooting. Connect when needed; it need not remain on the vehicle. The AGV industrial PC running the algorithm must remain powered and connected.

2.2 Operating Conditions and Limits

Check camera, computing hardware and processing time



Algorithm processing time

2D model-based recognition		3D geometry matching	
RK3576 / RK3588	**200 ms max.**	S11 + external compute	**200 ms max.**
M4 Mega onboard	**500 ms max.**	M4 Mega onboard	**200 ms max.**

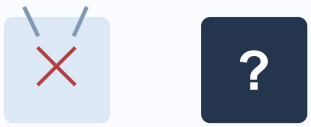
Select the range for the camera. The usable range varies with pallet size, type, material and model; validate it with site pallets.

Linux deployment is supported; confirm system and package compatibility. External-platform processing time depends on CPU, GPU and other resource usage. The figures exclude acquisition, communication and vehicle control. Validate the complete cycle on site; not for real-time motion feedback.

Deployment	Before use
S11 · 2D	Prioritize customer AGV compute using RK3576/RK3588, with compatible Linux and algorithm software. Requires site RGB training data. If capacity is insufficient, a board can be added by agreement after assessment, at extra cost. Not in the standard international offering.
S11 · 3D	Prioritize the customer’ s Linux AGV PC. Intel Core i5-class or RK3568 and above is recommended; confirm compatibility. If capacity is insufficient, a board can be added by agreement after assessment, at extra cost. Recommended internationally.
M4 Mega	Pure 3D uses onboard compute. No external PC required. 2D is slower and outside the standard international offering.

Operating Conditions and Limits

Pallet material



Reflective surfaces are unsuitable.
Assess black or low-reflectance pallets.

Pallet structure and visibility



Show the full entry face, legs and beams.
Occlusion and missing parts affect positioning.
Thin-legged, deformed or self-built pallets require assessment before use.

2D model suitability



Train with RGB images of site pallets.
Coverage depends on training and validation.

Entry-face angle



Within 15° relative to the camera.
Angles over 15° are filtered.

3D recognition height



Camera-to-floor height must match calibration.
A height change requires ground recalibration.

3D roll tilt



Roll tilt may prevent recognition.
Roll output does not mean all tilts are supported.

Safety Positioning results do not replace vehicle obstacle avoidance, emergency stops or safety interlocks.

Special Materials and Pallet Geometry

Check the material and fork-entry face image before assessing geometry. Black, bare-metal, thin-legged, deformed or custom-built pallets require Technical Support assessment.

Material	Imaging effect and action
Black / low reflectivity	Weak returns can leave gaps in leg point clouds. Assess reflectivity below 5% with particular care. With Technical Support, adjust integration time and the low-signal threshold, then check the leg image again.
Bare metal / highly reflective	Reflective surfaces are outside the standard operating conditions. Bare-metal pallets require separate assessment. Strong reflections can distort depth; an RGB image alone does not establish suitability.

Black-pallet assessment reference: lateral offset ±400 mm (camera optical center relative to pallet center), yaw ±10°. These are not guaranteed limits. Confirm usable distance against the camera range and actual imaging.

Geometry Reference for Assessment

The table gives reference dimensions for assessing pallet geometry. Validate suitability with site pallets. Use the camera recognition ranges given earlier in this section.

Item	Reference	Item	Reference
Pallet length / width	800–2000 mm	Pallet height	80–200 mm
Leg width	40–200 mm	Fork opening width	300–600 mm
Beam thickness	≥30 mm	Fork-entry face	Fully visible

The complete fork-entry face, legs and beam must be visible at the capture position. If forks, the vehicle body or cargo obscure them, adjust the capture position or mounting and validate again.



Occlusion example. Chinese labels: pallet at the center; forklift body in the red boxes.

Elevated Pickup: Reflector Placement

Navigation reflectors near a rack can interfere with depth imaging. Missing or distorted leg points may cause detection to fail. Keep reflectors away from pallet legs.

Placement recommendation: offset the reflector from the pallet legs by at least 400 mm.



Rack-mounted navigation reflector, outlined in green.

1. Check placement At the capture position, check reflector-to-leg separation at every planned rack level.

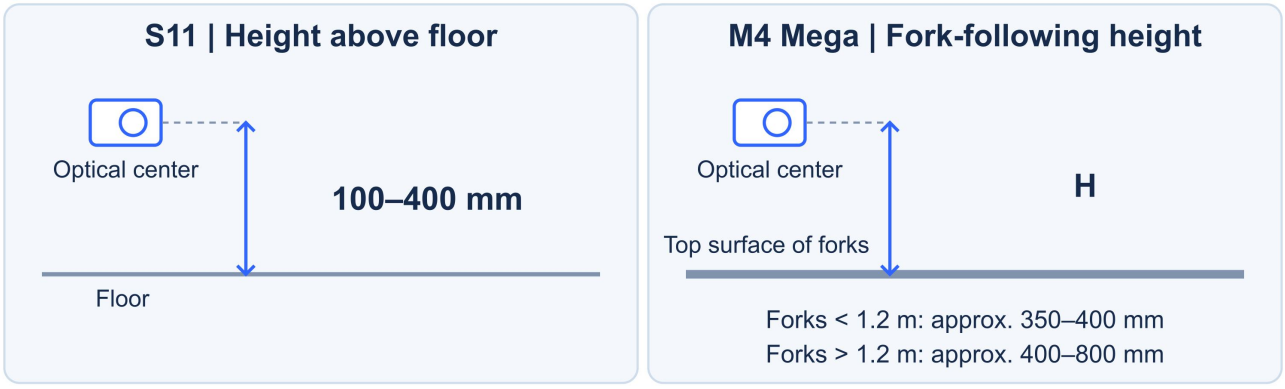
2. Check imaging Use AW3 depth images or point clouds to check that legs are complete and not visibly distorted. Confirm a valid positioning result.

3. Resolve interference Reposition the reflector first. If interference remains, contact Technical Support. Do not rely only on loosening template parameters.

See Section 2.3 for mounting and visibility requirements, and Section 3.4 for verification.

2.3 Installation Requirements

Use the height requirement for the camera: S11 is referenced to the floor; M4 Mega fork-following mounting is referenced to the fork top. Do not interchange these measurements.



Outside the range: confirm before installation

Confirm with Technical Support before installation

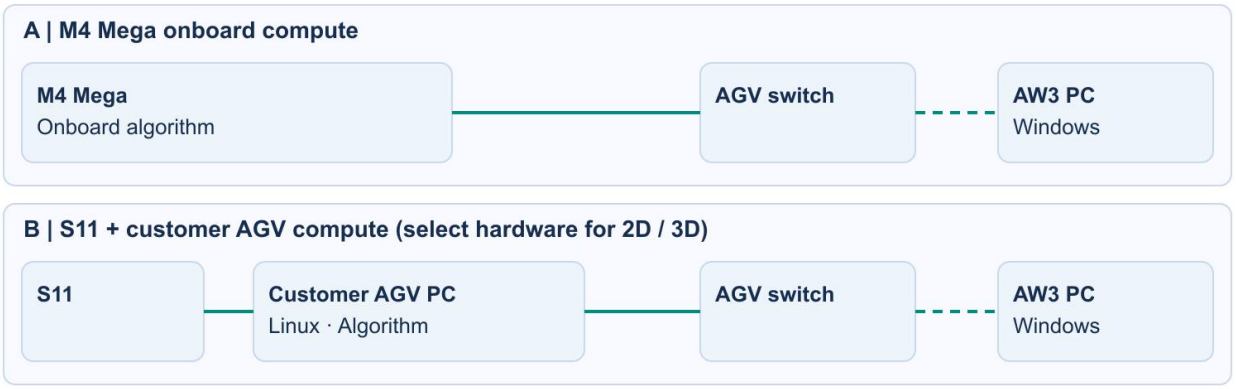
Not to scale. Apply each height range only with its specified camera and reference surface.

Check	Requirement
Center and level	Mount near the centerline between the two forks, within 50 mm laterally. Keep the camera level, its front vertical to the floor and pitch near zero.
Fixation and power	Secure brackets and cables against pulling, pinching and impact during mast travel. Follow the hardware manual; check power, network, streaming, and external compute-unit fixation and cooling.
S11 height above floor	100–400 mm, measured vertically from the camera optical center to the floor. Outside this range, confirm with Technical Support before installation.
M4 Mega fork-following height	Referenced to the fork top: forks < 1.2 m, approx. 350–400 mm; forks > 1.2 m, approx. 400–800 mm. Confirm the exact reference and mounting dimensions with Technical Support before installation. Confirm 1.2 m forks separately.
Fork-following mount	Fix the camera above the forks so it moves with them. Select its position using the height requirement for that camera.
Drop-down capture	Mount below the forks using a fixture; M4 Mega fork-following heights do not apply. Confirm fixture dimensions and capture position with Technical Support. Detect after full drop and camera stabilization.
Visibility	At nearest, near, far and rack capture positions, show the full entry face, legs and beam without fork/fixture occlusion. Recheck images and valid clouds after height changes.

Confirm uncertain mounting layouts, fixture dimensions or visibility with Technical Support before installation. Recalibrate after camera movement, bracket replacement or orientation changes.

Hardware Connections and AW3 PC

M4 Mega layout: pure 3D. Select S11 compute for 2D or 3D. Solid links stay connected; the dashed AW3 PC link is used as needed.



Both S11 2D and pure 3D prioritize the customer’ s AGV computing platform; see Section 2.2. No extra board is needed if requirements are met. If capacity is insufficient, a board can be added after assessment and agreement, at extra cost. Keep the algorithm host running and connected.

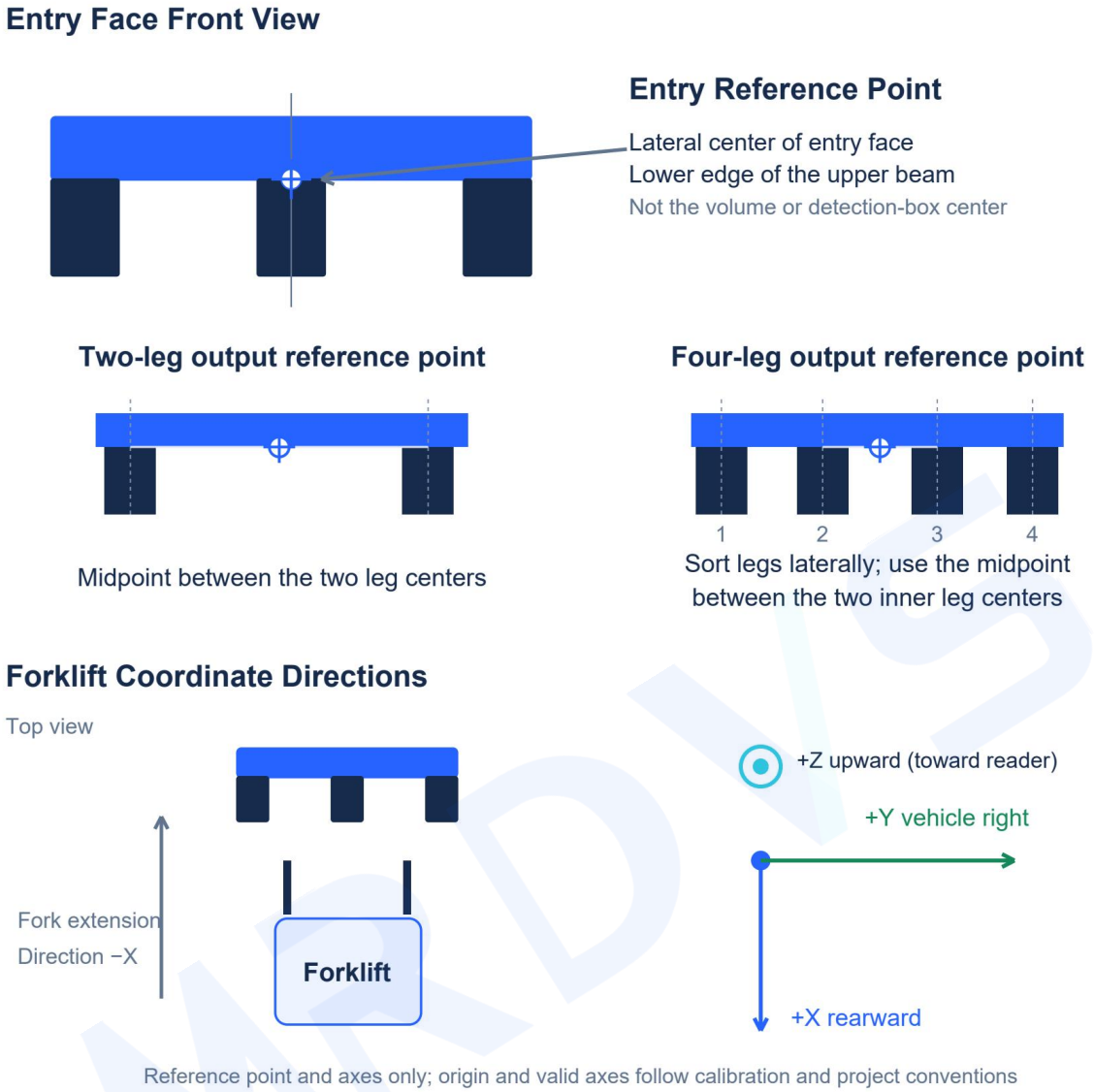
1. Connect the camera, algorithm host and AGV switch as shown. For deployment on the AGV industrial PC, install the algorithm package on that PC.

2. Connect a Windows PC to the AGV switch with an Ethernet cable and install AW3. Configure communicating network interfaces on the same subnet, using unique IP addresses.

3. Connect, configure and validate in AW3, then save and send the settings. Disconnect the PC after setup and reconnect for monitoring or troubleshooting. Keep the algorithm host connected and running.

See the AW3 User Manual for device access, network configuration and AW3 operations. Power connections are omitted; follow the hardware manuals for wiring and power.

2.4 Reference Point and Coordinates



Fork-entry reference point and forklift coordinate directions

1. The entry reference point lies on the lower edge of the upper beam at the entry face. Its lateral position is shown above; it is not the detection-box center or pallet volume center.
2. Forklift axes are shown in the diagram: +X rearward, +Y to the vehicle's right, and +Z upward. Position is in mm; angles are in °.
3. Do not mix camera and forklift coordinates. Agree on the reference point, axis directions and angle conventions before using results on the vehicle.
4. Verify directions with small fore/aft and lateral pallet movements, changing only one direction at a time.

Zero values and validity Negative values do not mean failure. Pitch is fixed at 0 as a placeholder, not a measured pitch angle.

3 Configuration and Calibration

3.1 Interface Overview

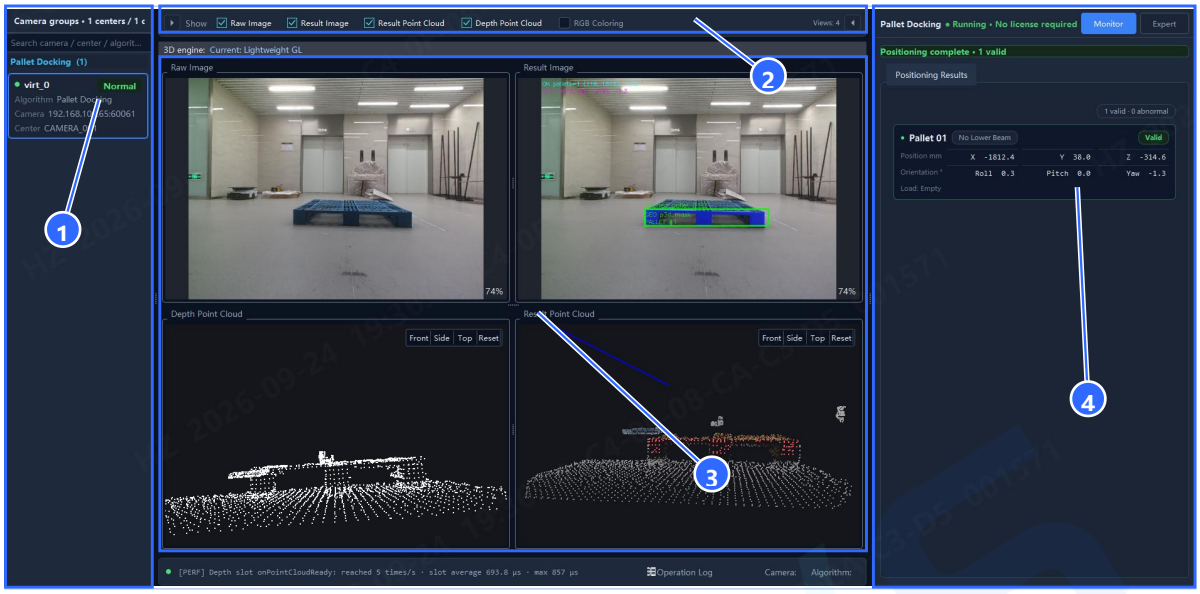


Figure 01 Main interface: ① Camera groups ② View controls ③ Images and point clouds ④ Algorithm panel

Area	Purpose
① Camera groups	Select [Pallet Docking]; check camera and center status.
② View controls	Select the image and point-cloud views to display.
③ Images and clouds	Inspect input and results. Front, Side, Top and Reset change the viewpoint only.
④ Algorithm panel	[Monitor] shows results; [Expert] provides positioning, detection and log export.

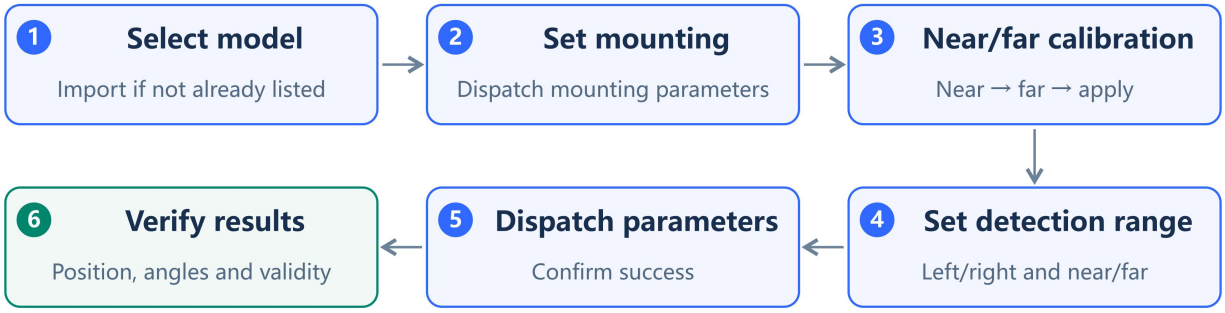
Views and Point Clouds

View	Purpose
Raw Image	Inspect targets, occlusion and lighting in the raw image.
Result Image	Check boxes or geometry against the actual target.
Depth Point Cloud	Check missing points, noise and target completeness.
Result Point Cloud	Compare fitted results with Depth Point Cloud.
RGB Coloring	RGB colors distinguish target/background; display only.

Dispatch after changing settings; save templates before dispatch. For calibration, apply the result and wait for completion. Viewpoint, zoom and RGB coloring affect display only.

3.2 2D Semantic Positioning

2D is not included in the standard international offering. Open [Expert] > [Positioning], switch to [2D], and confirm the current mode is [2D (Semantic)]. Follow the workflow below.



3.2.1 Import and Select a Model

Contact Technical Support for site-pallet RGB image requirements and the trained, validated project model. Import or select the model before installation settings and calibration.

1. In [Positioning], click [Select model...].

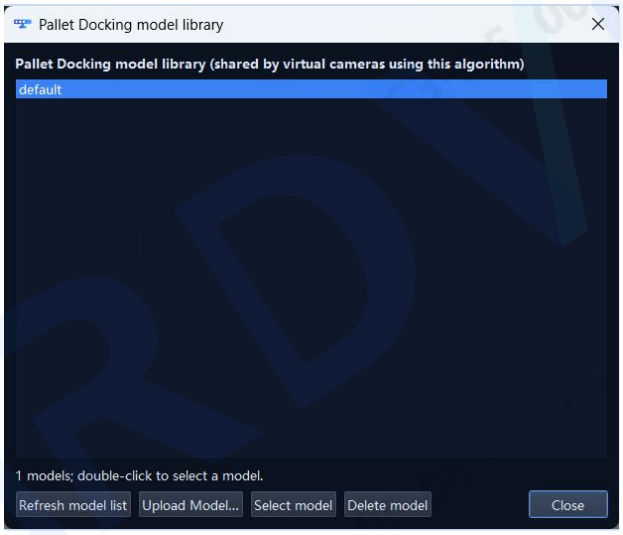


Figure 02 Pallet Docking model library

2. Upload if needed, refresh, then click [Select model] or double-click the model.

3. Check [Model Name] and click [Dispatch Parameters]. After success, continue with installation and calibration.

Button	Function
Refresh Model List	Refresh the available models.
Upload Model...	Import a model file supplied for the project.
Select Model	Activate the selected model; double-click also selects it.
Delete Model	Delete only when no other task uses the model.
Close	Close; dispatch parameters on return.

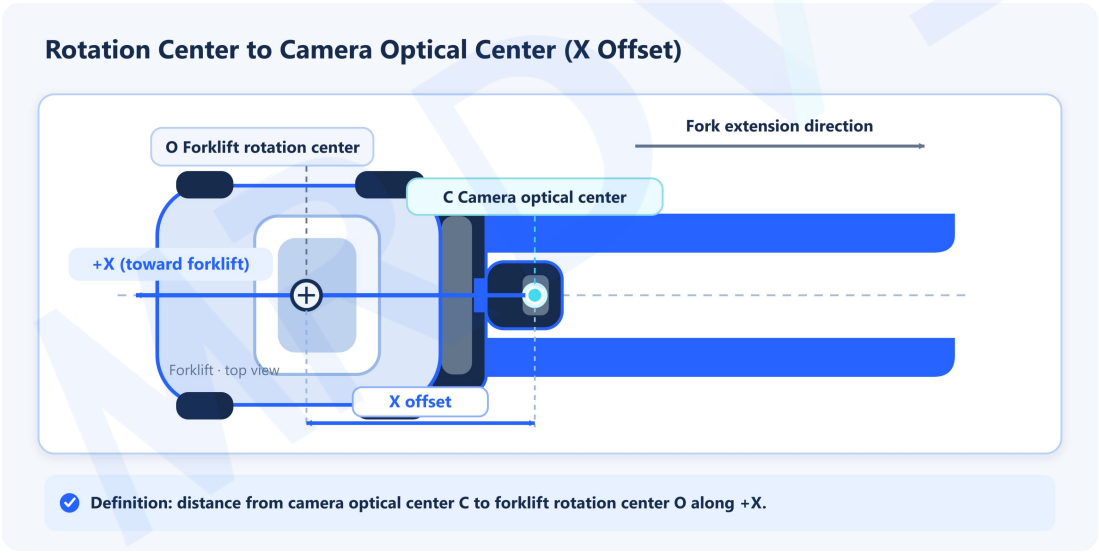
3.2.2 Installation and Calibration

Mounting Parameters

Open [Expert] > [Positioning] and set the following fields.

Parameter	How to set
Camera orientation	Select the actual mounting orientation: 0°, 90°, 180° or 270°.
Rotation-to-optical-center distance (mm)	Enter the fore/aft distance between the rotation center and camera optical center. Positive: rotation center behind the camera; negative: in front. Enter 0 for direct camera use without this compensation.
Fork insertion depth (mm)	Adjust the final entry-reference position. Leave at 0 if no adjustment is needed.
Fork angle (°)	Adjust the final docking angle. Leave at 0 if no adjustment is needed.

Click [Dispatch Parameters] and confirm success before calibration. Adjust insertion depth and fork angle during docking validation if needed, then dispatch again.



Measure the fore/aft distance between the rotation center and optical center

Ask the vehicle team to identify the rotation center; locate the optical center on the camera drawing. Recalibrate after changing orientation or this distance.

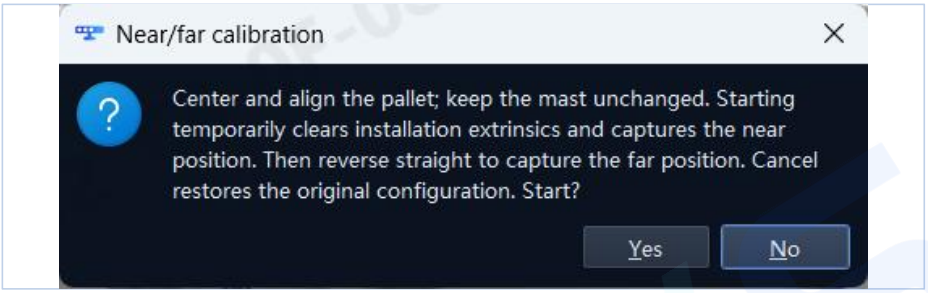
Near/far calibration corrects mounting angles and lateral offset. Set fore/aft mounting distance with the rotation-to-optical-center distance.

Near/Far Calibration

Place one pallet on level ground and face it squarely. For near calibration, choose a position with the full entry face visible, stable recognition and clear image/depth data. The minimum recognition distance depends on pallet size and type. Keep the pallet, mast height/tilt and vehicle heading unchanged.

1 Calibrate Near

Stop the vehicle. Click [Calibrate near], then [Yes] in the confirmation dialog. Continue only after near capture completes and [Calibrate far] becomes available.



Near-calibration confirmation: click [Yes] to start or [No] to return.

2 Calibrate Far

Reverse straight 400–800 mm without turning or moving sideways. Stop and click [Calibrate far]. Wait for capture to finish and [Apply calibration] to become available. Click it and wait for the green completion message. The result is saved and dispatched automatically.

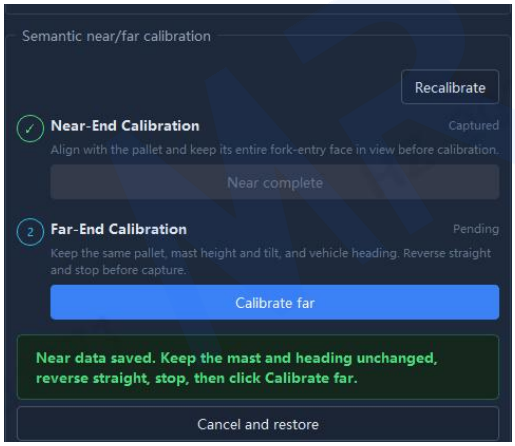


Figure 03 Near complete: ready for far capture

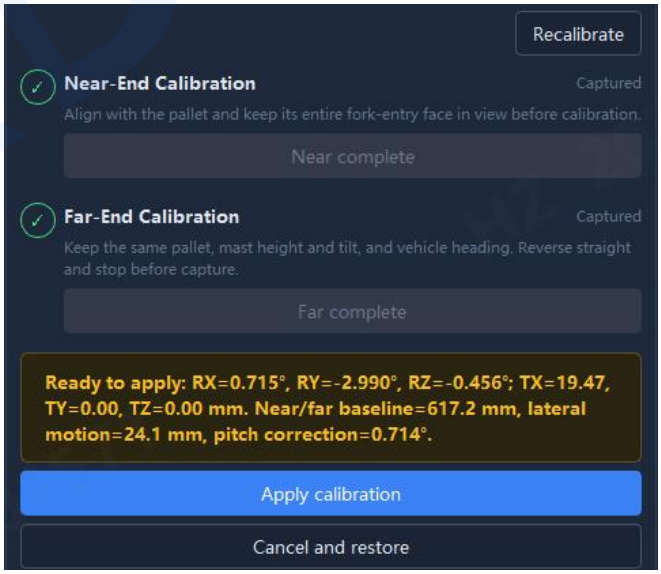


Figure 04 Far complete: ready to apply

If you exit or change the pallet, mast or vehicle heading, click [Cancel and restore] and start again from near capture.

3 Verify Results

Check that [Extrinsic Matrix] has updated. In [Monitor], check the same pallet at near, middle and far positions for position, angles and valid status.



Figure 05 Calibration complete and extrinsic matrix

Matrix fields: Pitch, Yaw and Roll correct mounting angles; Lateral is the left/right offset, Height the vertical offset, and Fore/aft the front/back offset. These are calibration parameters; Pitch in the positioning result remains fixed at 0.

If completion is not reported, do not proceed. Use Section 6.2 to troubleshoot the failed near or far capture.

3.2.3 Detection Range Settings

Set the left/right detection range and near/far distances, in mm.

Setting	Meaning and setup
Lateral range (mm)	Set the left and right detection limits relative to the camera.
Detection distance (mm)	Set the nearest and farthest detection distances from the camera.

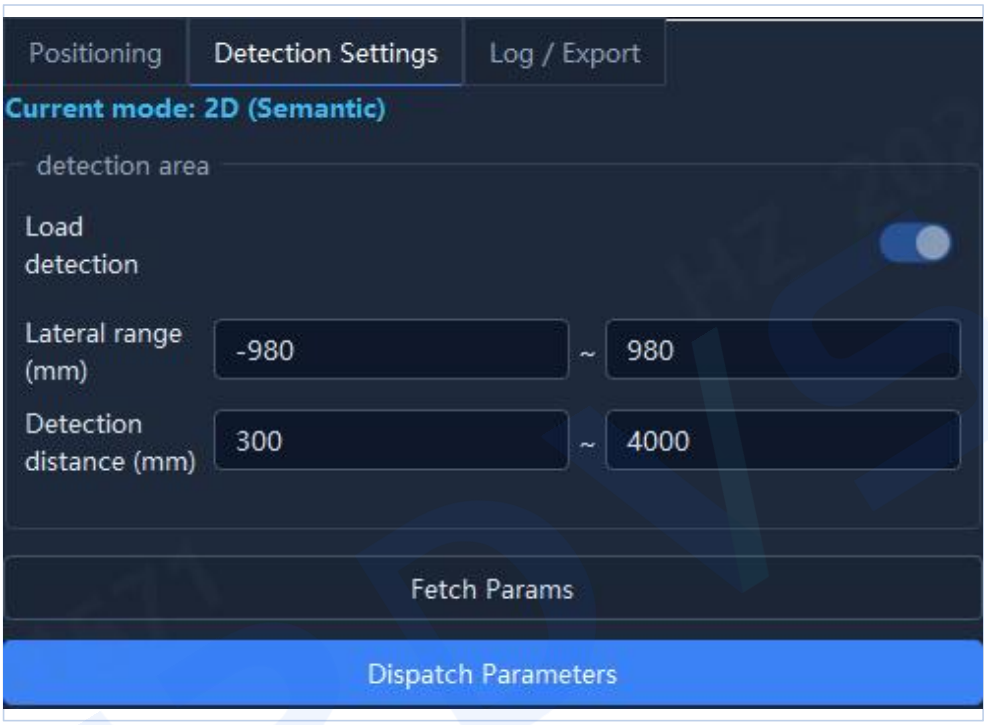


Figure 06 2D detection settings: lateral range, distance and parameter dispatch

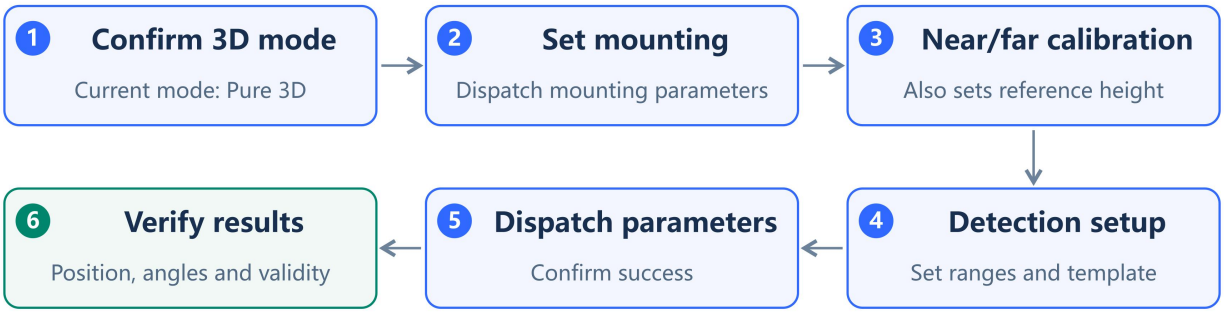
The Load detection control shown does not apply to 2D. Set only the lateral range and detection distance here.

1. Open [Expert] > [Detection Settings]. Click [Fetch Params] to view current settings.
2. Set lateral limits to cover the working width and exclude unwanted neighboring areas.
3. Set near and far limits, with the near value lower than the far value. 2D distance depends on the camera and site model; validate actual near and far positions.
4. Click [Dispatch Parameters] and confirm success. In [Monitor], check results inside, at and outside the limits.

2D functions 2D detects multiple eligible pallets in view without multi-pallet configuration. It does not support load detection or require a geometric pallet template.

3.3 Pure 3D Positioning

Open [Expert] > [Positioning], switch to [3D], and confirm the current mode is [3D (Pure 3D)]. Follow the workflow below.



Capture height

Before calibration, set the camera to its actual height above the floor during detection. Near/far calibration also sets the reference height automatically.

> **Keep the 3D detection height fixed**
> During detection, keep the camera-to-floor height the same as during floor-reference calibration. **After changing detection height, auto-calibrate the floor reference again, dispatch parameters, and verify positioning before use.**

A capture-height change alone does not require near/far recalibration. Repeat full calibration if camera orientation or its mounting relationship to the vehicle changes.

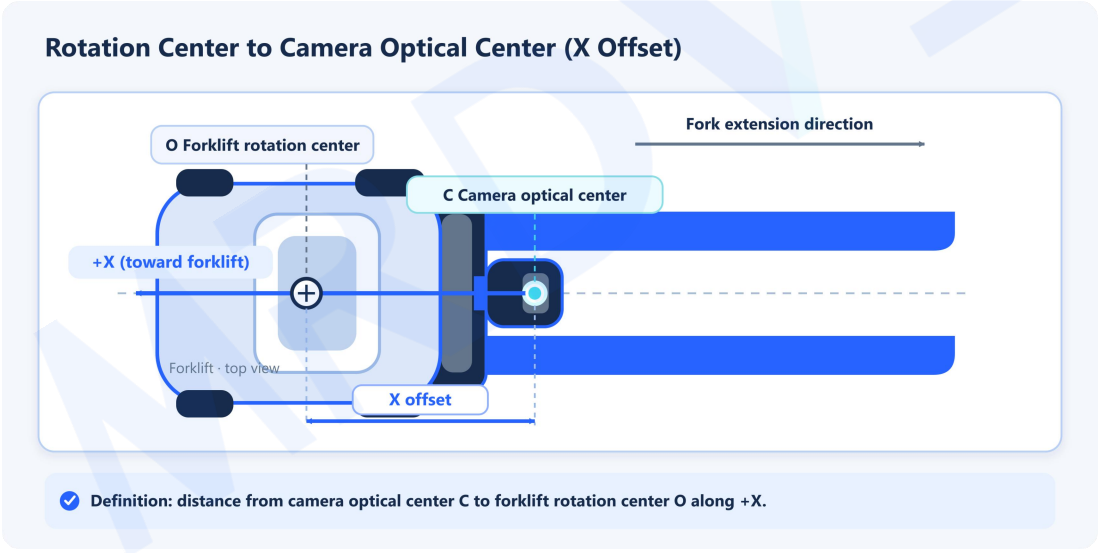
3.3.1 Installation and Calibration

Mounting Parameters

Open [Expert] > [Positioning] and set the following fields.

Parameter	How to set
Camera orientation	Select the actual mounting orientation: 0°, 90°, 180° or 270°.
Rotation-to-optical-center distance (mm)	Enter the fore/aft distance between the rotation center and camera optical center. Positive: rotation center behind the camera; negative: in front. Enter 0 for direct camera use without this compensation.
Fork insertion depth (mm)	Adjust the final entry-reference position. Leave at 0 if no adjustment is needed.
Fork angle (°)	Adjust the final docking angle. Leave at 0 if no adjustment is needed.

Click [Dispatch Parameters] and confirm success before calibration. Adjust insertion depth and fork angle during docking validation if needed, then dispatch again.



Measure the fore/aft distance between the rotation center and optical center

Ask the vehicle team to identify the rotation center; locate the optical center on the camera drawing. Recalibrate after changing orientation or this distance.

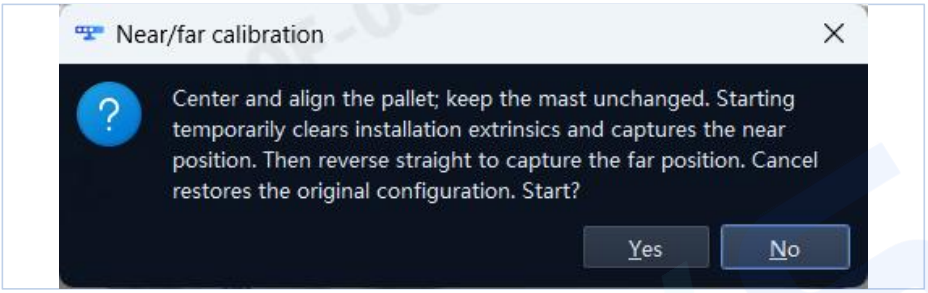
Near/far calibration corrects mounting angles and lateral offset. Set fore/aft mounting distance with the rotation-to-optical-center distance. Pure 3D also calibrates reference height.

Near/Far Calibration

Use a three-leg pallet. Keep the floor and beam edges visible. Set the operating camera height and face the pallet squarely. For near calibration, choose a position with the full entry face visible and stable recognition. The minimum recognition distance depends on pallet size and type. Keep the pallet, mast height/tilt and vehicle heading unchanged.

1 Calibrate Near

Stop the vehicle. Click [Calibrate near], then [Yes] in the confirmation dialog. Continue only after near capture completes and [Calibrate far] becomes available.



Near-calibration confirmation: click [Yes] to start or [No] to return.

2 Calibrate Far

Reverse straight 400–800 mm without turning or moving sideways. Stop and click [Calibrate far]. Wait for capture to finish and [Apply calibration] to become available. Click it and wait for the green completion message. The result is saved and dispatched automatically.

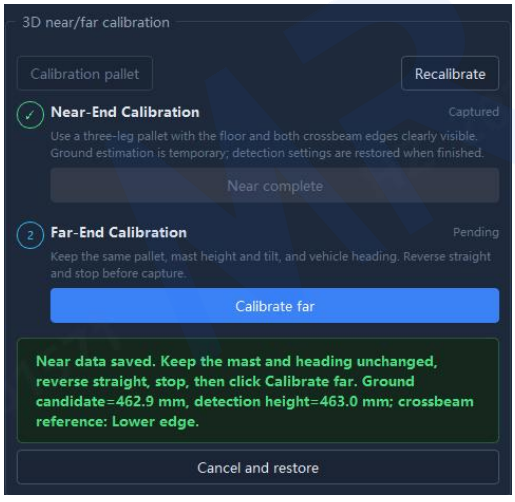


Figure 07 Near complete: ready for far capture

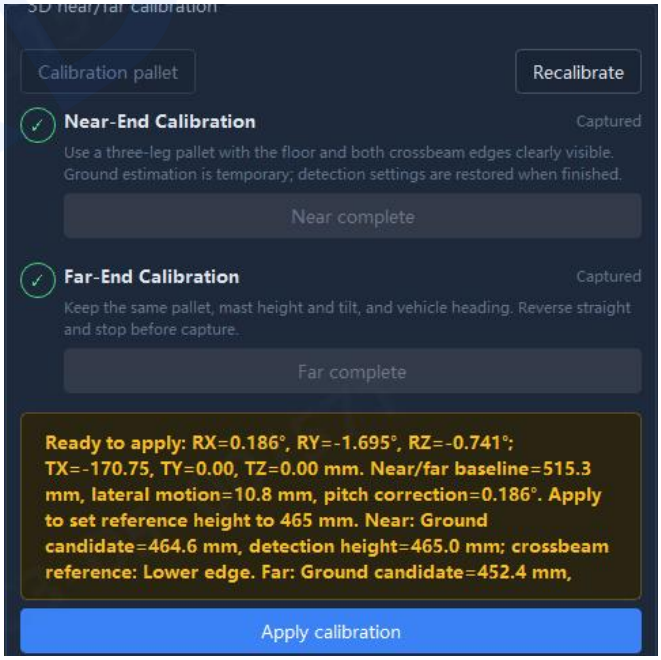


Figure 08 Far complete: ready to apply

If you exit or change the pallet, mast or vehicle heading, click [Cancel and restore] and start again from near capture.

3 Verify Results

Check that [Extrinsic Matrix] and [Reference Height] have updated. In [Monitor], verify positioning at near, middle and far distances. Use the calibrated height, not the screenshot value.



Figure 09 3D calibration complete and extrinsic matrix

Figures 08 and 09 are separate calibration examples; their values differ.

If completion is not reported, do not proceed. Use Section 6.2 to troubleshoot the failed near or far capture.

Matrix fields: Pitch, Yaw and Roll correct mounting angles; Lateral is the left/right offset, Height the vertical offset, and Fore/aft the front/back offset. These are calibration parameters; Pitch in the positioning result remains fixed at 0.

Adjust the Calibration Pallet Only If Needed

If default settings do not recognize the pallet, first check full visibility and floor depth data, then compare measured dimensions. To adjust, click [Calibration pallet] in [Positioning]. Calibration requires a three-leg pallet.

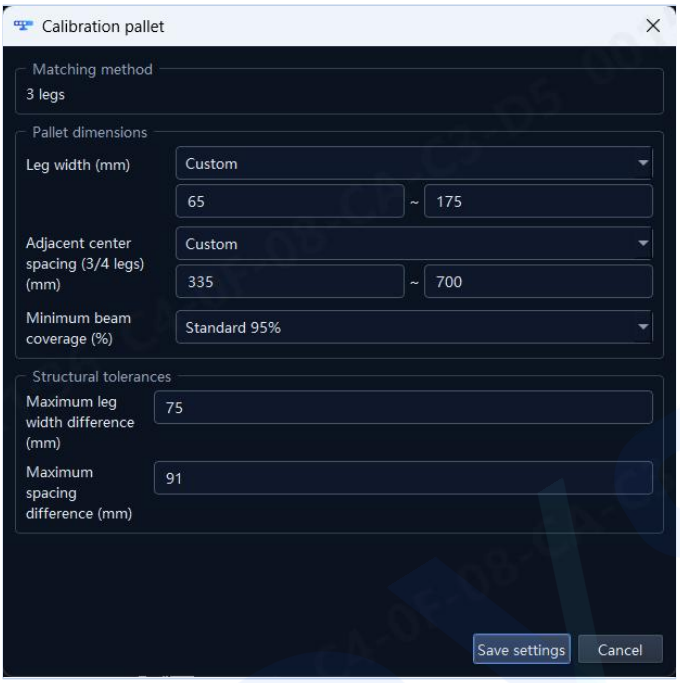


Figure 10 Calibration pallet settings

Field	How to set
Leg width (mm)	Enter minimum and maximum visible width of one leg.
Adjacent center spacing (mm)	Set the allowed left–middle and middle–right center spacing, not the clear fork-opening width.
Minimum beam coverage (%)	Checks the beam between the legs. At 95%, at least 95% of horizontal sample positions must contain valid beam points.
Maximum leg width difference (mm)	Limits width differences between matched legs.
Maximum spacing difference (mm)	Limits center-spacing differences in the matched structure.

Select a matching preset or enter measured ranges. Click [Save settings], return to the positioning page and click [Dispatch Parameters]. Restart near/far calibration from the near capture.

3.3.2 Detection Page and Detection Area

Complete calibration first, then configure detection settings in screen order. If capture height changes, recalibrate the floor using the steps on the next page.

1. Open [Detection Settings] and click [Fetch Params] to view current settings.

2. Select a matching [Current Template]. If none fits, add one as described in Section 3.3.3, then return here.

3. Configure the detection settings listed below.

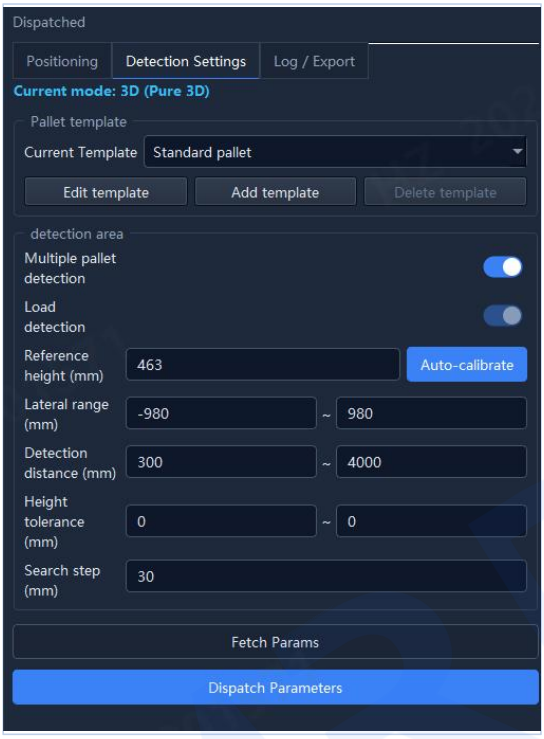


Figure 11 Pure 3D detection settings

Setting	Meaning and operation
Multiple pallet detection	Detects stacked pallets. Check each layer for misses or duplicates.
Load detection	Indicates whether a pallet carries a load, not its weight, volume or stability.
Reference height (mm)	Detection reference height, obtained by calibration.
Lateral range (mm)	Set the left and right detection limits relative to the camera.
Detection distance (mm)	Set the nearest and farthest detection distances from the camera.
Height tolerance (mm)	Set how far to search above and below the reference height.
Search step (mm)	Vertical distance between successive search heights.

Save and Verify

1. Click [Dispatch Parameters] and confirm success.

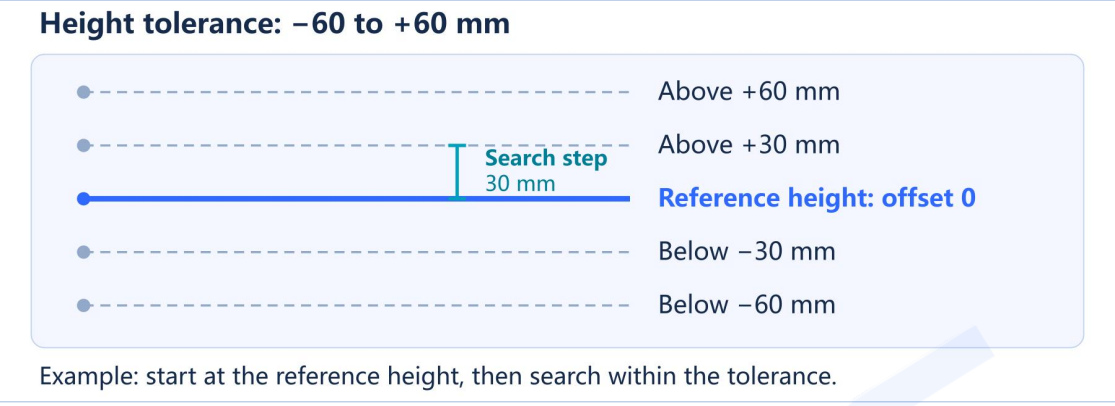
2. Open [Monitor] and test one pallet. Check that its status is [Valid] and its position and angles match the actual pallet.

3. For stacked pallets, enable [Multiple pallet detection] and dispatch again. Compare the results with the actual layers for misses or duplicates.

Reference Height, Height Tolerance and Search Step

Keep the 3D detection height fixed

During detection, keep the camera-to-floor height the same as during floor-reference calibration. **After changing detection height, auto-calibrate the floor reference again, dispatch parameters, and verify positioning before use.**



Parameter	Purpose and setting
Reference height	Floor reference for detection. Near/far calibration sets it; [Auto-calibrate] updates it separately.
Height tolerance	Search range below and above the reference. Both limits at 0 mean detection at the reference height only.
Search step	Interval between search heights. For example, 30 mm means one attempt every 30 mm.
Detection mode	**Search rule**
Single pallet	Try the reference height first, then step above and below within the tolerance. Stop when a pallet is found.
Multiple pallets	Does not use the height tolerance or search step on this page.

Recalibrate the Floor after Changing Capture Height

1. Fix the camera at its actual detection height, stop the vehicle and check that the floor point cloud is valid.

2. In [Detection Settings], click [Auto-calibrate] beside [Reference Height] and wait for the value to update.



3. Click [Dispatch Parameters]. After success, verify pallet positioning in [Monitor].

If camera orientation or its mounting relationship to the vehicle changes, repeat near/far calibration.

3.3.3 Detection Templates and Parameters

Add, Edit and Delete Templates

1. Open [Expert] > [Detection Settings]. Click [Add Template] and enter [Template Name]. To edit an existing template, select it first and click [Edit Template].

Configure one common template to cover the pallets in the project. Use matching presets for leg width, center spacing and beam coverage, or [Custom] ranges based on measurements. Only when pallets differ too much to share a template should you create another and switch manually. Parameters follow screen order.



Figure 12 Edit a detection template

2. Select the actual leg count and set dimensions and structural tolerances. The following pages explain the parameters.

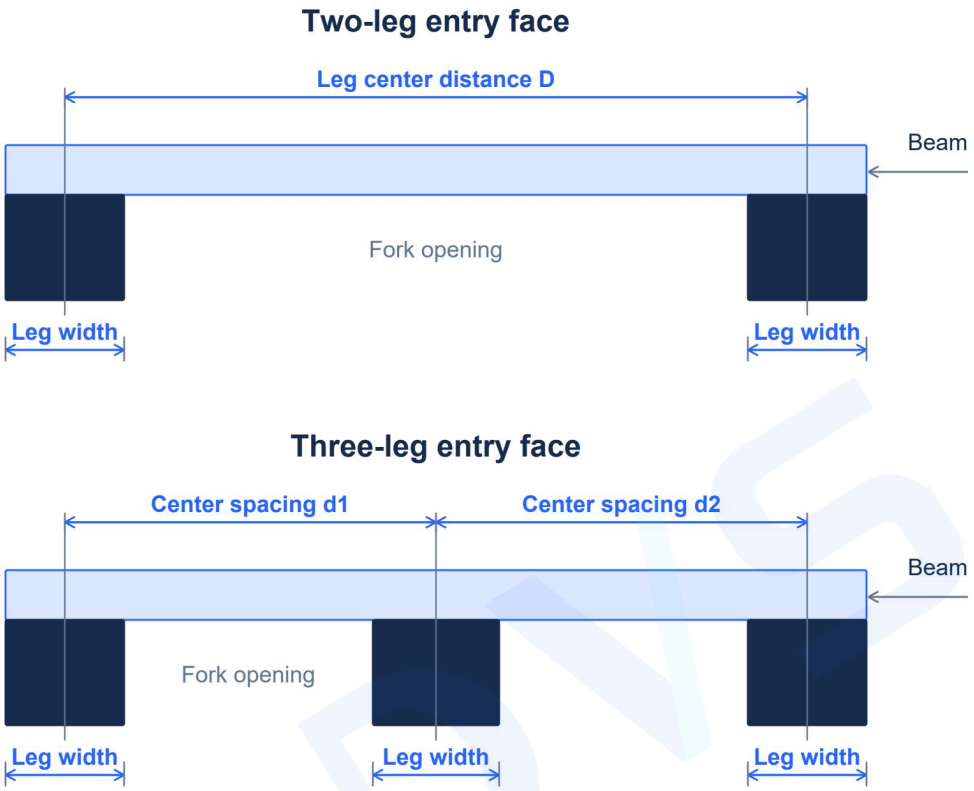
3. For a new template, click [Save template]; for an existing one, click [Save parameters].

4. Return to the detection page. Check [Current Template], click [Dispatch Parameters], and verify with actual pallets.

To delete a custom template, select it in [Current Template], click [Delete Template], check its name and confirm. Then select the template to use and dispatch parameters. Built-in templates cannot be deleted.

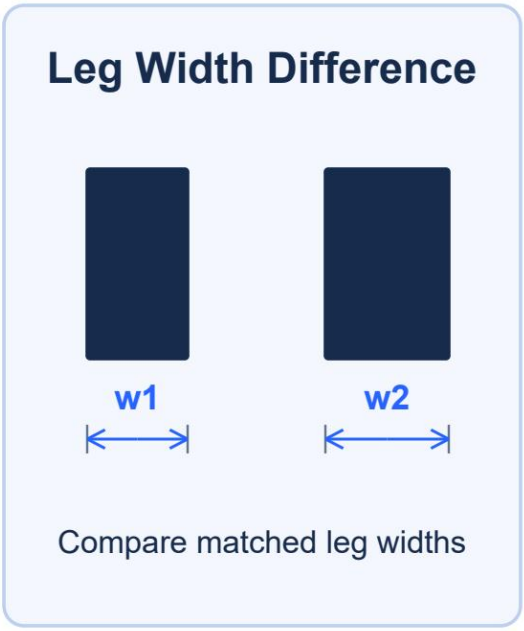
Dimensions

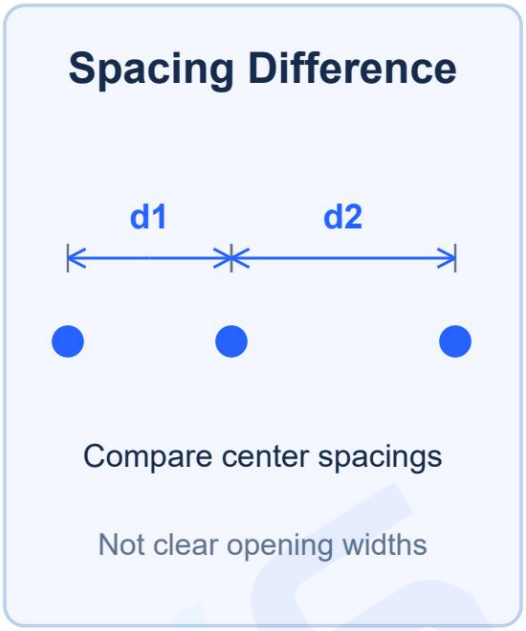
Open [Add Template] or [Edit Template] in [Detection Settings] and complete the fields below. Measure spacing between leg centers, not the clear fork opening.



Parameter	How to set
Leg count matching	Select 2, 3 or 4 legs to match the fork-entry face. Multiple selections are allowed; select at least one.
Leg width (mm)	Enter the minimum and maximum visible width of one leg, not the full pallet width.
Adjacent center spacing (3/4 legs)	Set the range between adjacent leg centers. Four-leg matching uses the upper limit.
Leg center distance (2 legs) (mm)	Set the distance range between the two leg centers. Default: 700–1600 mm.
Minimum beam coverage (%)	Checks the beam between the legs. At 95%, valid beam points must be found at least 95% of the horizontal sample positions. Lower it for thin or interrupted beams, then check for false pallet detections in the background.

Structural Tolerances





Parameter	How to set
Maximum leg width difference (mm)	Set the largest permitted width difference between matched legs.
Maximum spacing difference (mm)	Set the largest permitted difference between matched center spacings.

Enter structural tolerances as numeric values. Change one parameter group at a time, save and dispatch, then check for missed pallets and false background detections.

3.4 Monitor and Verify Positioning Results

After dispatch succeeds, open [Monitor]. Check that images keep updating and detection boxes identify the intended pallets. Then read the result cards.

2D Semantic Monitor Page

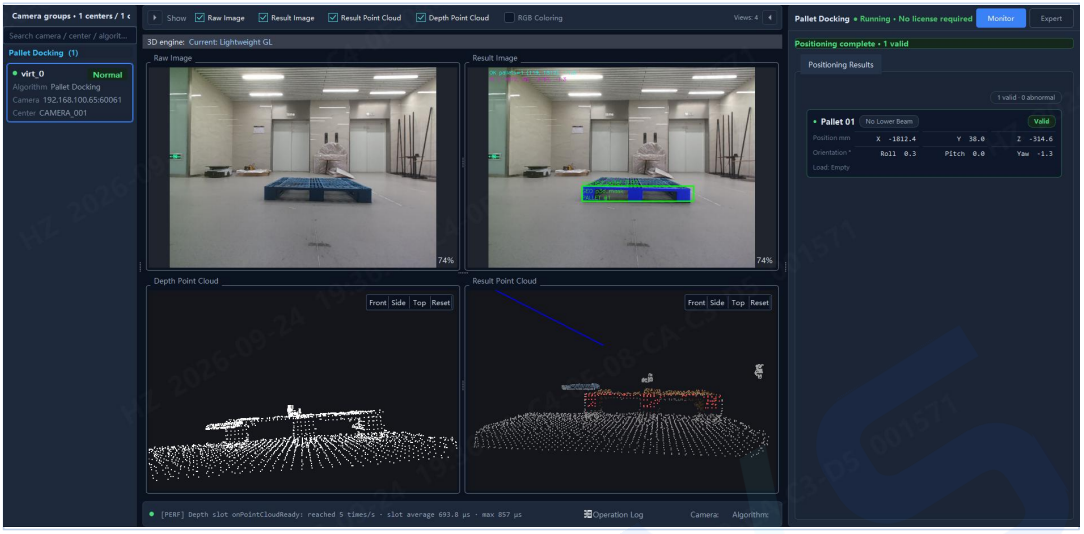


Figure 13 2D Monitor: semantic results, point clouds and per-pallet result cards

Pure 3D Positioning Monitor

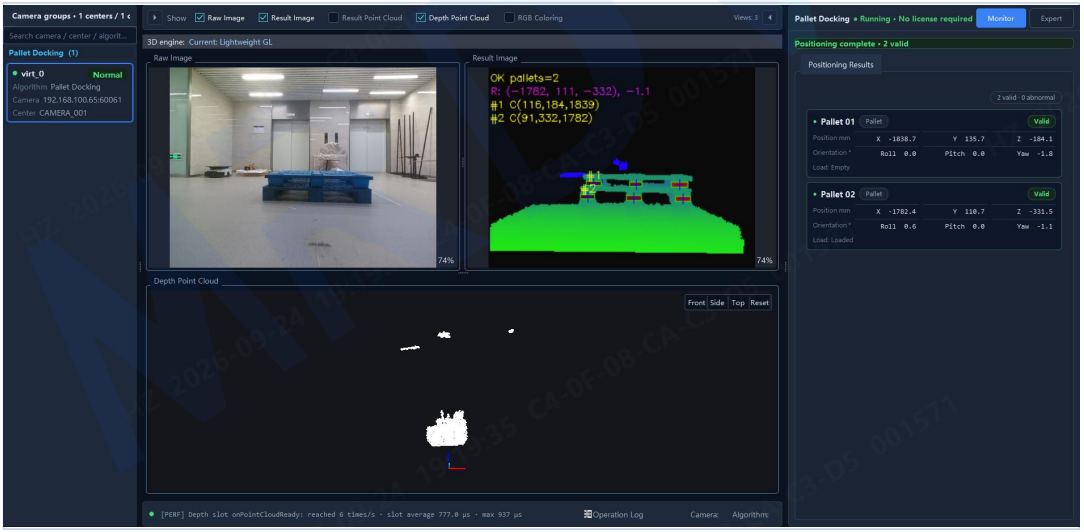


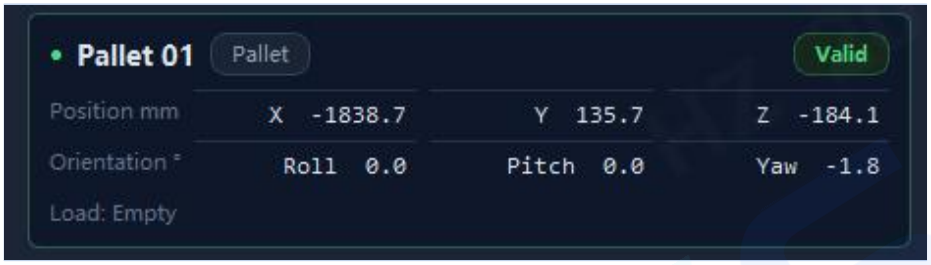
Figure 14 3D monitor: point-cloud matching and per-pallet result cards

3.4.1 Read Status and Result Cards

2D result card: class, validity, position and orientation



3D result card: validity, position and orientation; load status when enabled



Item	Description
Valid / abnormal count	Counts current positioning results. Inspect individual cards when errors occur.
Green valid status	The pallet has a valid positioning result. Still confirm it is the correct target.
Yellow abnormal status	Read the error message. Do not use a pose whose position or orientation is shown as “—”.
Position X / Y / Z	Entry reference position in forklift coordinates, in mm.
Roll / Pitch / Yaw	Valid orientation outputs are Roll and Yaw, in °. Pitch is fixed at 0 and is not a valid output.
Load	Use only in 3D with load detection enabled. In 2D, this field does not indicate whether the pallet carries a load.

Status distinction “Positioning complete” means processing has ended, not that every target was positioned successfully.

Card numbers identify the current results. They are not persistent IDs or picking priorities.

3.4.2 Invalid Positioning Results

Do not use these results for docking. Check positioning status even when a detection box is present.

Result	Meaning	Action
Positioning Error	No valid pose. The example shows an entry-face angle over 15° relative to the camera.	Realign the pallet to reduce its angle relative to the camera, then detect again.
Incomplete Structure	In 2D, occlusion prevents recognition of the complete entry face.	Remove the obstruction, show the full entry face and detect again.



Figure 15 Detection box with pallet rotation

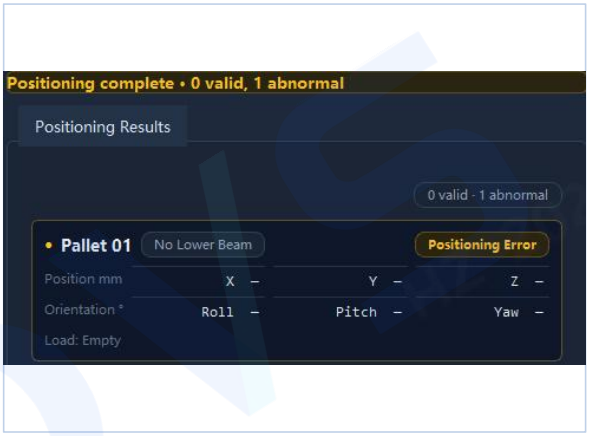


Figure 16 Positioning Error: pose unavailable

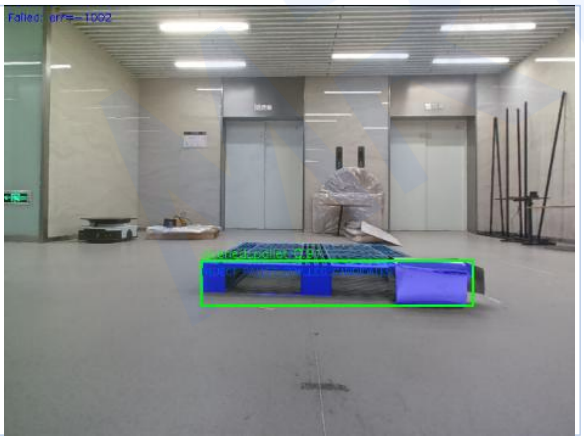


Figure 17 Occluded entry-face structure

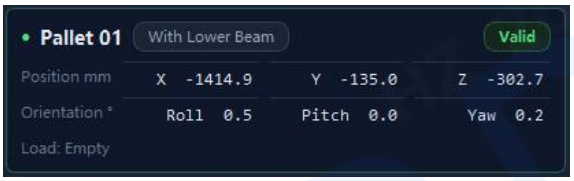
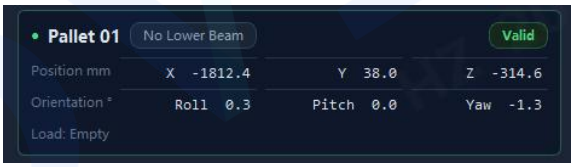


Figure 18 Incomplete Structure: pose unavailable

3.4.3 2D Pallet Structure Classification

The standard international offering uses pure 3D geometric positioning. It does not include 2D semantic positioning. In 2D mode, the model detects whether the fork-entry face has a lower crossbeam. Classification is automatic with a suitable model; no separate switch is required.

In [Monitor], compare the class label on the pallet result card with the actual fork-entry face.

With Lower Beam **Closed pallet**	**No Lower Beam** **Open pallet**
	
A continuous crossbeam joins the legs below the openings.	No continuous crossbeam below the fork openings.
	

Pallet structure and matching result labels

Here, Closed pallet corresponds to [With Lower Beam], and Open pallet to [No Lower Beam]. The lower beam is the bottom board spanning beneath the fork openings.

Classification describes only the entry face seen by the camera, not the whole pallet base. Different entry faces of the same pallet may differ. Only 2D provides this classification; check positioning status separately.

2D mode does not support load detection. Ignore the [Load] field in these screenshots.

3.4.4 Verify Positioning and Docking

Test scenario	What to check
Near, middle and far	Stable results for the same pallet, without clear distance-related drift in lateral position or angle.
Lateral and fore/aft motion	Result direction and magnitude match the coordinate convention.
Different angles	Reported orientation matches the pallet and meets project tolerances.
Stacked pallets	No missing or duplicate layers; the vehicle selects the correct target layer.
No target and occlusion	The vehicle stops using old results when the target is removed or the pose is invalid.
Load status	Test empty and loaded pallets only in 3D with load detection enabled. Do not use this judgment in 2D.
External communication	Target, status and pose agree between sender and receiver.

If Deviations Occur

1. Check input views, target correspondence, mode and model or template.
2. Check mounting orientation and calibration conditions; recalibrate if needed.
3. Adjust insertion depth and angle only after mounting geometry is correct. Do not use offsets to compensate for invalid calibration.

Record versions, model/template, calibration settings and results. Verify accuracy and repeatability against project criteria. Section 2.2 gives algorithm time; set vehicle waits and timeouts from the measured complete cycle, with margin. Validate each mode used.

3.5 Daily Operation

1. Check the lens, bracket and cables before work; confirm the mount is secure.
2. Start AW3, select the correct camera group, and check continuous streaming and normal algorithm operation.
3. Use a known pallet to check position, direction and validity.
4. Confirm vehicle-side result reception and safety interlocks before automatic operation.

Stop conditions Stop using a target pose if the camera is offline, frames freeze, positioning fails, the target is wrong or results expire. Recheck after recovery.

4 Using the Functions

This chapter covers pallet template management, multiple-pallet detection and load detection.

4.1 Manage Pallet Templates

Use one common template for the project wherever possible. Create another and switch manually only when pallet differences prevent sharing. Templates can be added, edited and deleted; see Section 3.3.3.

4.2 Multiple Pallet Detection

Use this function when multiple pallets are present in one view. 2D needs no multi-pallet switch. In 3D, enable [Multiple pallet detection] to search stacked pallets layer by layer.

1. 2D: select a suitable model and inspect each pallet's result. No leg-count or other geometric setup is needed.
2. 3D: open [Expert] > [Detection Settings], enable [Multiple pallet detection], click [Dispatch Parameters].
3. In [Monitor], compare actual pallet count and layers. Check misses, duplicates and target correspondence.

Missing layers: check ranges, occlusion and point clouds. Duplicates: check overly broad templates and false structure matches. Match each result to a physical layer. The vehicle selects the target and motion; do not use list order or card numbers.

4.3 Load Detection

Pure 3D only: [Load detection] checks whether the pallet carries a load.

1. Confirm stable pure 3D positioning and a template that matches the pallet.
2. Open [Expert] > [Detection Settings], enable [Load detection], and click [Dispatch Parameters].
3. Test empty and loaded pallets separately; check load status on the [Monitor] result card.
4. Revalidate after changing goods or occlusion. If results disagree with the scene, retain images, clouds and logs.

Condition	Note
2D mode	Load detection is not supported. Do not use the displayed load field to judge empty or loaded.
Disabled or unknown status	Do not treat this as empty.
Positioning error	Restore valid pallet positioning before using load status.
Use of result	Indicates load presence, not weight, volume or load stability.

5 Communication and Result Data

Pallet docking supports TCP, UDP, CAN, RS-485 and API access. Select an interface, then follow its setup and decoding instructions below.

Interface	Use and reference
TCP / UDP	Send start/stop commands and receive binary results. See 5.4.
RS-485	Control detection and read results using Modbus RTU. See 5.5.
CAN	Exchange control and result frames. See 5.6.
API access	Read pallet results via the SDK. Not supported for the 2D model-based solution. See 5.7.

JSON and binary messages use different formats. Sections 5.2–5.3 describe AW3 JSON; 5.4–5.7 describe binary and SDK interfaces. Output fields differ between interfaces. Do not use JSON field order to decode binary bytes.

5.1 Configure and Verify Communication

Configure triggering and result forwarding in AW3 Protocol Setup. See Section 7.4 of the AW3 User Manual for the UI steps. Configure binary ports, RS-485, CAN and SDK access as described in the respective sections below.

Use a known pallet to compare status, position, angles and units at the receiver. Test no-target conditions, disconnection and recovery. Do not use invalid, timed-out or stale results for docking.

5.2 Read Positioning Results

Check overall validity and the pallet status before reading position and angles. pallets[] refers to each returned pallet.

Result	Field	Meaning
Overall validity	valid	true means the result is valid.
Pallet count	count	Number of returned pallets.
Positioning status	pallets[].ret_code	0 means this pallet was positioned successfully.
Error description	pallets[].error_str	Failure reason; empty on success.
X position	pallets[].forklift_pose.x	mm; positive points backward.
Y position	pallets[].forklift_pose.y	mm; positive points to the vehicle’ s right.
Z position	pallets[].forklift_pose.z	mm; positive points upward.
Roll angle	pallets[].forklift_pose.roll	Roll in degrees.
Yaw angle	pallets[].forklift_pose.yaw	Yaw in degrees.

Pitch is fixed at 0 and is not a valid measured output.

These fields apply to AW3 JSON only. Binary TCP/UDP, CAN, RS-485 and SDK fields, scaling and height definitions are described below. Do not assume that each interface provides all five JSON pose axes.

5.3 Result JSON Example

Main fields from a successful 2D positioning result.


```
{
  "valid": true,
  "ret_code": 0,
  "error_str": "",
  "count": 1,
  "pallets": [
    {
      "pallet_type": "opened_pallet",
      "ret_code": 0,
      "error_str": "",
      "forklift_pose": {
        "x": -1304.7950439453125,
        "y": -9.663567543029785,
        "z": -320.1379699707031,
        "yaw": 1.6952972412109375,
        "pitch": 0.0,
        "roll": 0.4580188989639282
      }
    }
  ]
}
```


The example is successful and returns one Open pallet with no lower beam (opened_pallet). Forklift coordinates: X = −1304.795 mm, Y = −9.664 mm, Z = −320.138 mm; Roll = 0.458°, Yaw = 1.695° (rounded to three decimals here).

Class • 2D only	pallets[].pallet_type: opened_pallet means No Lower Beam; closed_pallet means With Lower Beam. Do not use this field for lower-beam classification in 3D.
Load • 3D only	pallets[].pallet.is_onload: with load detection enabled and valid positioning, true means loaded and false means empty. Disabled, missing or null is not empty. Do not use this field in 2D.

5.4 TCP and UDP Hexadecimal Messages

The device acts as the TCP server on default port 5501. The default UDP port is 8000. Connect to the device IP and select HEX sending mode. Detection must use the always-on operating mode.

Send start and stop commands

Action	Bytes to send (HEX)
Start detection	60 04 00 00
Stop detection	60 04 00 01

60 is the command marker; 04 is the total command length of 4 bytes. The final two bytes are the command code: 0000 starts and 0001 stops detection. Send four bytes, not the text string “60 04 00 00”.

Decode a response

Byte position[1]	Field	Decoding
1–2	Return code	uint16; 0 = success. Discard pose on any other value.
3–4	Frame length	uint16; use the received length to identify a complete frame.
5–8	Signs	X, Y, Yaw, height, in order; 00 = positive, 01 = negative.
9–12	Absolute X	uint32; divide by 1000 for mm.
13–16	Absolute Y	uint32; divide by 1000 for mm.
17–20	Absolute Yaw	uint32; divide by 1000 for degrees.
21–24[2]	Absolute height	Present in height-bearing messages; divide by 1000 for mm.
25–32[2]	Extension	Read only when the corresponding extension is present; not extra pose axes.

Hexadecimal decoding example

[1] Byte positions start at 1. Multibyte integers are big-endian: most significant byte first.

[2] Read trailing fields only when present in the applicable message version. Do not replace missing fields with zero. TCP reads may contain partial or multiple frames.

This response has 20 bytes. Its length field is 00 14, or decimal 20. It contains X, Y and Yaw, without height.

```
00 00 00 14 01 00 00 00
00 17 AA FC 00 00 18 38 00 00 2A 41
```

Bytes	Meaning	Decoded value
00 00	Return code	0: positioning succeeded
00 14	Length	20 bytes
01 00 00 00	Signs	X negative; Y and Yaw positive
00 17 AA FC	Absolute X	1,551,100 ÷ 1000 = 1551.1 mm
00 00 18 38	Absolute Y	6,200 ÷ 1000 = 6.2 mm
00 00 2A 41	Absolute Yaw	10,817 ÷ 1000 = 10.817°

Result: X = −1551.1 mm, Y = +6.2 mm, Yaw = +10.817°. This frame contains neither height nor Roll.

Check the return code and frame length before applying values, signs and scaling. Compatibility-protocol height is measured from the optical center, positive downward. AW3 JSON Z uses the calibrated coordinate system; the two are not interchangeable.

5.5 RS-485 Communication

Use Modbus RTU. Defaults: 9600 bit/s, 8 data bits, no parity, 1 stop bit (8N1). The default slave address is 0x17; the configurable range is 1–247.

Action	Request (HEX)
Start detection	17 06 00 78 00 01 CA E5
Stop detection	17 06 00 78 00 00 0B 25
Read the first four result fields	17 04 00 7A 00 08 D2 E3

Request field	Meaning
17	Slave address 0x17 (decimal 23).
06 / 04	06: write one register; 04: read input registers.
00 78 / 00 7A	0x0078 controls detection; results start at 0x007A.
00 01 / 00 00	Control value: 1 = start, 0 = stop.
00 08	Read 8 registers, totaling 16 data bytes.
Last two bytes	Modbus CRC16, low byte sent first.

Response example and field order

The response below contains 16 data bytes: X, Y, Yaw and the return code, four bytes each. Raw integers are 1, 2, 3 and −2. Return code −2 indicates failure; do not use the first three values as a valid pose.

```
17 04 10
00 00 00 01
00 00 00 02
00 00 00 03
FF FF FF FE
86 D2
```

Decode result fields as signed 32-bit integers, with negative values in two’ s complement. Field order: X, Y, Yaw, return code, height H. Each occupies 2 registers (4 bytes). The example reads the first four fields, totaling 8 registers; it does not include height H.

5.6 CAN Communication

Use the CANopen interface at 250 kbit/s, with 11-bit standard frame IDs and 8 data bytes per frame. The COB-IDs below follow the pallet protocol.

Action	COB-ID	8 data bytes (HEX)
Start detection	0x212	40 00 01 00 01 00 00 00
Stop detection	0x212	40 00 01 00 00 00 00 00

Data[4] controls detection: 01 starts and 00 stops it. Send all other bytes exactly as shown.

Result COB-ID	Data bytes (zero-based)	Fields
0x192	0–1 / 2–3 / 4–7	count / result / Yaw
0x292	0–3 / 4–7	X / Y
0x392	0–3 / 4–7	Height / LeftHoleWidth
0x492	0–3 / 4–7	RightHoleWidth / MiddleLegWidth

count is the query counter. Check result before using the associated result frames. X, Y, Yaw, Height and width fields are transmitted as actual value × 1000; divide by 1000 after decoding. Position and width use mm; angles use degrees.

Read CAN results using the field layout above, without the separate TCP/UDP sign bytes. Do not combine frames from different updates. This interface has no separate Roll field.

5.7 API Access

The SDK API in this section does not support the 2D model-based solution.

Use the SDK application_pallet.cpp example. After connecting and starting the stream, refresh the data before reading LX_PTR_ALGORITHM_OUTPUT. The result type is LxPalletPose.

Call sequence: DcOpenDevice → DcSetIntValue (MODE_PALLET_LOCATE) → DcStartStream → refresh and read results → DcStopStream → DcCloseDevice.

Read one result (handle is connected and streaming; inside a function returning bool):

```
void* data = nullptr;
if (DcSetCmd(handle, LX_CMD_GET_NEW_FRAME) != LX_SUCCESS)
    return false;
if (DcGetPtrValue(handle, LX_PTR_ALGORITHM_OUTPUT,
                  &data) != LX_SUCCESS || data == nullptr)
    return false;
const auto* pose = static_cast<const LxPalletPose*>(data);
if (pose->return_val != 0) return false;
const float x = pose->x, y = pose->y, yaw = pose->yaw;
return true;
```

Structure field	Meaning
return_val	0 = positioning succeeded; otherwise discard the pose.
x / y / yaw	Single-pallet X, Y and Yaw as floats; not TCP byte fields.
extents[0]	Height from the optical center, positive downward; not JSON Z.
extents[10]	Valid pallet count n; up to 4 in this SDK extension format.
extents[11 + 4i … 14 + 4i]	Pallet i: X, Y, Yaw×1000, height; i starts at 0 and is less than n.

Multipallet extensions are ordered from top to bottom; the vehicle selects its target. Check return codes and null pointers, and read or copy data within the same frame. The SDK owns the returned memory; do not free it. For five-axis pose, 2D class and 3D load fields, use the JSON definitions in 5.2–5.3. LxPalletPose does not contain every JSON field.

6 Troubleshooting and Maintenance

First check that Raw Image and Depth Point Cloud keep updating. Restore streaming if input fails; otherwise investigate positioning performance.

6.1 System and Streaming Errors

Symptom	Checks and action
No image or frozen frames	Check camera power, cable, network and central-service status. Confirm the selected camera group.
Camera online, no valid depth	Check distance, reflections, lens contamination and cloud quality. Adjust capture conditions per the hardware manual.
Algorithm stopped or license error	Check algorithm status, version and license messages. Follow the AW3 User Manual.
Streaming fails after camera setup	If LxCameraViewer was just used, open [Device Connection / Base Camera] > [Central Service] > [Restart current center].
AW3 has results, receiver does not	Check protocol, address, port, receiving service and camera group. Verify as in Section 5.1.

After Recovery

1. Confirm Raw Image and Depth Point Cloud update continuously rather than show an old frame.
2. Check positioning with a known pallet.
3. Confirm the receiver gets new results after recovery before resuming work.

Operating rule Do not continue fork insertion with an old pose after disconnection, frozen frames or an algorithm error.

6.2 Positioning Performance Issues

See Section 3.4.2 for invalid-result meanings and examples. Use this table to find the corrective action.

Message or issue	Action
Positioning Error	Check the specific error first. If the angle is too large, realign the pallet to within 15° of the camera and detect again.
Incomplete Structure	Remove occlusion, make the full entry face visible, and detect again.
No pallet recognized in 2D	Confirm model support; check the image, depth and detection range.
No pallet recognized in 3D	Check leg count, dimensions, beam coverage, reference height and range. Reposition visibly tilted pallets.
Reflective or black pallet	Reflective surfaces are unsuitable. Assess black or low-reflectance pallets on site; check point-cloud completeness, not RGB appearance alone.
Background matched in 3D	Narrow the detection area and tighten template dimensions and tolerances.
Missing or duplicate pallets	Check occlusion and clouds. In 3D, also check multi-pallet detection and template.
Near calibration failed	Stop and face the pallet squarely. Check the 2D model, or the 3D three-leg settings and floor cloud.
Far calibration failed	Complete near capture first. Reverse straight 400–800 mm, stop and capture; keep mast position and heading fixed.
Unreasonable calibration or position drift	Check mounting orientation and rotation-center distance. Cancel and restore, realign and recalibrate.
Incorrect load state	Confirm load detection is enabled and positioning is valid. Clear occlusion, check clouds and retain misclassified samples.
Parameters not applied	Confirm dispatch succeeded. Fetch parameters and check camera group, mode and template.

Troubleshooting method Change one parameter group at a time. For recurring issues, save settings and corresponding data, then export as in Section 6.3.

6.3 Export Troubleshooting Data

1. Open [Expert] > [Log / Export]. Record fault time and the full message; click [Copy All] if needed.

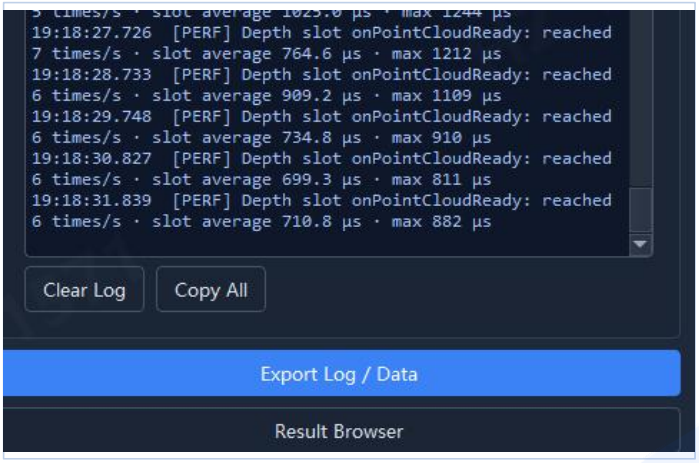


Figure 19 Log and export controls

2. Click [Export Log / Data]. Follow Chapter 8 of the AW3 User Manual to select algorithm, category and a time range covering before and after the fault.

3. Use [Result Browser] to check result images and JSON. Confirm export completeness before sending to Technical Support.

Do not clear logs before export. Include versions, configuration, site photos, pallet dimensions and distance, failure time and reproduction steps. Verify the exported images, clouds and logs.

6.4 Maintenance and Revalidation

When	Requirement
Daily maintenance	Check lens, bracket, fasteners, cables, power and cooling. Clean per the hardware manual without moving the mounted camera.
Camera or mounting changes	Recheck orientation and camera-to-fork geometry. Complete near/far calibration.
3D capture height changes	Recalibrate the floor reference height with [Auto-calibrate], click [Dispatch Parameters], then verify pallet positioning.
Model or software update	Back up settings, check versions and the model report, then test representative on-site pallets.
Different pallet type	2D: check model coverage. 3D: configure a matching template.
Scene or mast conditions change	Check calibration still applies. Verify depth quality, ranges, position and direction.

Keep records Retain validated models, templates and settings for post-maintenance comparison and recovery.

Revision History and Technical Support

Version	Date	Revision
V0.1	2026-09-28	V0.1: consolidated repeated workflows, monitoring and parameter guidance; retained full functionality and troubleshooting; aligned 2/3/4-leg rules and the 400–800 mm calibration reverse distance.
V0.2	2026-10-30	Corrected the interface list to TCP, UDP, CAN, RS-485 and API; added binary fields, decoding examples, serial/CAN commands and SDK access; updated the communication reference.

Technical Support

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Scan to open MRDVS on GitHub